

# AN ELEMENTARY COMPARISON FOR FOUR-ARC REGIONS IN A PLANAR STRIP DENSITY

Proof built, Lean-verified, and semantically accepted by  
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**Abstract.** For the planar density equal to one on a horizontal strip and to a constant  $\lambda > 1$  outside it, every member of the explicit symmetric four-arc family has a three-arc competitor of exactly the same weighted area and strictly smaller weighted perimeter. We prove the quantitative improvement  $\gamma_\lambda/h$ , including the curvature endpoint  $h = 1$ . The argument compares  $P - hA$  at equal curvature, then selects the greatest equal-area root. Coordinate constructions and integral identities justify the geometric comparison independently of any classification of minimizers.

## 1. INTRODUCTION AND STATEMENT

Cañete, Miranda Jr. and Vittone [1, §3.2] studied the isoperimetric problem for a plane whose density jumps across the boundary of a strip. Their Lemma 3.8 and Proposition 3.9 reduce the possible minimizers, up to density-preserving isometries, to four families with vertical reflection symmetry: disks contained in the strip, stadiums with straight sides on the interfaces, three-arc regions with a bottom segment, and four-arc regions with congruent exterior caps. Their Conjecture 3.12 predicts the exclusion of the last family. We give the corrected elementary comparison of the supplied note by Skilyn Leon, Rosa Pavlak, Evelyn Pulla and Xi Sisi Shen [2]. The comparison is self-contained for its explicit regions; its implication for the unrestricted conjecture uses the external classification and is discussed in Section 6.

Fix a real number  $\lambda > 1$ . The density and the geometric quantities used throughout are

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1, \end{cases} \quad (1.1)$$

$$A_\lambda(F) = \int_F f_\lambda \, dx \, dy, \quad P_\lambda(F) = \int_{\partial F} f_\lambda \, d\mathcal{H}^1.$$

Here  $\mathcal{H}^1$  is Euclidean one-dimensional Hausdorff measure and  $\partial F$  is the complete topological boundary. In particular, the density is one on both interfaces  $|y| = 1$ . The regions constructed below are closed and bounded, and both integrals are finite.

LEAN REFERENCES

`CMVGeometry.lean` StripDensity; WeightedArea; FrontierMeasure; WeightedPerimeter

Write

$$\gamma_\lambda = \lambda \arccos(1/\lambda) - \sqrt{1 - \lambda^{-2}}. \quad (1.2)$$

The four-arc region  $C_h$  has common circular radius  $h^{-1}$ , where  $0 < h \leq 1$ , and exterior half-angle  $\alpha = \arccos(h/\lambda)$ . Thus  $\lambda \cos \alpha = h$ . Its exact coordinate definition, and that of the three-arc family  $E_s$ , are given in Section 2.

LEAN REFERENCES

`CMVElementary.lean` CMVElementary.gamma

`CMVElementaryGeometry.lean` CMVElementary.GeometryBridge.fourCandidate\_snell

**Theorem 1 (geometric comparison).** *For every real  $\lambda > 1$ , one has  $\gamma_\lambda > 0$ . For every real  $h$  with  $0 < h \leq 1$ , there exists a real  $r$  with  $0 < r < h$  such that*

$$\begin{aligned} A_\lambda(E_r) &= A_\lambda(C_h), \\ P_\lambda(C_h) - P_\lambda(E_r) &> \frac{\gamma_\lambda}{h} > 0. \end{aligned} \quad (1.3)$$

*Consequently,  $C_h$  does not minimize weighted perimeter at its weighted area.*

LEAN REFERENCES

`CMVElementaryGeometry.lean` CMVElementary.geometric\_comparison

In both statements of Theorem 1 in [2], including its display (3.2), the expression on the left must be  $P_\lambda(C_h) - P_\lambda(E_r)$ , as in (1.3), not  $A_\lambda(C_h) - P_\lambda(E_r)$ . The latter is false. The original note is retained unchanged as a source and style reference; its literal erroneous displays are not the result proved here.

## 2. COORDINATE REGIONS AND GEOMETRIC FORMULAS

For  $-1 \leq z \leq 1$ , set

$$\begin{aligned} B(z) &= \lambda \arccos(z/\lambda) + \arcsin z, \\ D(z) &= \sqrt{1 - z^2/\lambda^2} - \sqrt{1 - z^2}. \end{aligned} \quad (2.1)$$

The dependence on  $\lambda$  is suppressed. Notice that  $D(z) \geq 0$ : the first radicand is at least the second. For  $0 < s \leq 1$ , put  $q = 2s - 1$  and define

$$\begin{aligned} A_3(s) &= \frac{B(q) + \pi/2 + (2s+1)D(q)}{s^2}, & P_3(s) &= \frac{2(B(q) + \pi/2 + D(q))}{s}, \\ A_4(s) &= \frac{2(B(s) + sD(s))}{s^2}, & P_4(s) &= \frac{4B(s)}{s}. \end{aligned} \quad (2.2)$$

The values  $A_3(1)$  and  $P_3(1)$  are scalar endpoint values only; the geometric competitors have parameter strictly less than one.

LEAN REFERENCES

`CMVElementary.lean` `CMVElementary.B`; `CMVElementary.D`; `CMVElementary.D_nonneg`; `CMVElementary.A3`;  
`CMVElementary.P3`; `CMVElementary.A4`; `CMVElementary.P4`

For  $0 < h \leq 1$ , let  $R = h^{-1}$ ,  $\alpha = \arccos(h/\lambda)$  and  $d = D(h)/h$ . Define

$$\begin{aligned} K_h &= \{(x, y): -1 \leq y \leq 1, |x| \leq d + \sqrt{R^2 - y^2}\}, \\ H_h &= \{(x, y): y \geq 1, x^2 + (y - (1 - R\cos\alpha))^2 \leq R^2\}, \\ \sigma(x, y) &= (x, -y), \quad C_h = K_h \cup H_h \cup \sigma(H_h). \end{aligned} \quad (2.3)$$

Reflection  $\sigma$  supplies the lower cap. Its non-endpoint exterior arc lies in  $y < -1$ , not merely in  $y < 1$ . Both  $C_h$  and the three-arc region below are symmetric about the vertical axis;  $C_h$  additionally has horizontal reflection symmetry.

LEAN REFERENCES

`CMVElementaryGeometry.lean` `CMVElementary.K`; `CMVElementary.upperCap4`; `CMVElementary.sigma`;  
`CMVElementary.C`

For  $0 < s < 1$ , now let  $q = 2s - 1$ ,  $R = s^{-1}$ ,  $\alpha = \arccos(q/\lambda)$ , and  $b = D(q)/s$ . Define

$$\begin{aligned} K_s^{(3)} &= \{(x, y): -1 \leq y \leq 1, |x| \leq b + \sqrt{R^2 - (y - (R - 1))^2}\}, \\ H_s^{(3)} &= \{(x, y): y \geq 1, x^2 + (y - (1 - R\cos\alpha))^2 \leq R^2\}, \\ E_s &= K_s^{(3)} \cup H_s^{(3)}. \end{aligned} \quad (2.4)$$

Because  $R > 1$ , the strip radicand is nonnegative. Since  $b \geq 0$ , every horizontal section is an interval. The cap is defined as a disk intersected with a half-plane, so its definition does not discard the major-segment branch.

LEAN REFERENCES

`CMVElementaryGeometry.lean` `CMVElementary.K3`; `CMVElementary.upperCap3`; `CMVElementary.E`;  
`CMVElementary.GeometryBridge.K3_core`

**Proposition 1 (geometric realization).** *For every  $\lambda > 1$ , the regions  $C_h$ ,  $0 < h \leq 1$ , and  $E_s$ ,  $0 < s < 1$ , are closed, bounded and measurable. Their densities are integrable over the regions and their complete frontiers, whose Euclidean lengths are finite. They satisfy*

$$\begin{aligned} A_\lambda(C_h) &= A_4(h), \quad P_\lambda(C_h) = P_4(h), \\ A_\lambda(E_s) &= A_3(s), \quad P_\lambda(E_s) = P_3(s). \end{aligned} \quad (2.5)$$

*The open attaching chords are interior to the unions. The bottom frontier of  $E_s$  is exactly the segment  $\{(x, -1): |x| \leq b\}$ . For  $s < 1/2$ , its exterior cap has half-angle greater than  $\pi/2$ ; for  $s = 1/2$ , the region is the disk of radius two centered at  $(0, 1)$ .*

LEAN REFERENCES

`CMVElementaryGeometry.lean` `CMVElementary.geometric_realization`

*Proof.* At  $y = \pm 1$ , the half-width in (2.3) is

$$d + \sqrt{R^2 - 1} = \frac{D(h) + \sqrt{1 - h^2}}{h} = \frac{\sin \alpha}{h}.$$

Thus the cap and core chords agree. At a strict interior point of either chord the disk inequalities are strict on the appropriate sides of the interface, giving a neighborhood inside their union. The exposed frontier consists precisely of the two strip arcs and the two exterior arcs. Their intersections are confined to the four junctions and hence have zero  $\mathcal{H}^1$ -measure. The strip arcs have total length  $4\arcsin(h)/h$ ; the exterior arcs have total length  $4\alpha/h$ . It follows that

$$P_\lambda(C_h) = \frac{4\arcsin h + 4\lambda\alpha}{h} = P_4(h). \quad (2.6)$$

These are lengths of the actual frontier, not independently assigned boundary data.

LEAN REFERENCES

`CMVElementaryGeometry.lean` `CMVElementary.GeometryBridge.four_upper_chord`;  
`CMVElementary.GeometryBridge.four_lower_chord`

`CMVGeometry.lean` `FourArcAssembly.frontier_carrier`

`FrontierPerimeter.lean` `fourArc_frontier_weightedPerimeter_eq`

Horizontal integration gives the central area and the area of one cap:

$$\begin{aligned} |K_h| &= \frac{4D(h)}{h} + \frac{2\sqrt{1-h^2}}{h} + \frac{2\arcsin h}{h^2}, \\ |H_h| &= \frac{\alpha - \sin \alpha \cos \alpha}{h^2}. \end{aligned} \quad (2.7)$$

The overlaps are contained in the interface lines and have zero planar area. Adding the central area and twice the weighted cap area, then using  $\lambda \cos \alpha = h$ , gives  $A_\lambda(C_h) = A_4(h)$ . The formulas remain valid at  $h = 1$ .

LEAN REFERENCES

`CMVGeometry.lean` `FourArcAssembly.weightedArea_formula`

`FourArcCandidate.lean` `FourArcCandidate.candidate_area_formula`

`CMVElementaryGeometry.lean` `CMVElementary.GeometryBridge.C_area`

For (2.4), the upper half-width is

$$b + \frac{\sqrt{1-q^2}}{s} = \frac{\sin \alpha}{s}.$$

The strict attaching chord is again interior to the union. At  $y = -1$ , the radicand vanishes; the whole segment of length  $2b$  remains exposed and has density one. Each of the two strip arcs has central angle  $\arcsin q + \pi/2$ , and the exterior arc has central angle  $2\alpha$ . Therefore

$$P_\lambda(E_s) = \frac{2(\arcsin q + \pi/2) + 2\lambda\alpha}{s} + 2b = P_3(s). \quad (2.8)$$

No attaching chord contributes to this perimeter.

LEAN REFERENCES

```
CMVElementaryGeometry.lean CMVElementary.GeometryBridge.three_strict_chord;
                             CMVElementary.GeometryBridge.E_bottom_frontier
TypeThreeAssembly.lean TypeThreeAssembly.frontier_carrier;
                       TypeThreeAssembly.weightedPerimeter_formula
```

The corresponding area calculation is

$$\begin{aligned} |K_s^{(3)}| &= \frac{4D(q)}{s} + \frac{\arcsin q + \pi/2 + q\sqrt{1-q^2}}{s^2}, \\ \lambda|H_s^{(3)}| &= \frac{\lambda\alpha - qs\sin\alpha}{s^2}. \end{aligned} \quad (2.9)$$

Their sum is  $A_3(s)$ . The cap formula follows by integrating the disk sections and holds throughout  $0 < \alpha < \pi$ , including  $\alpha > \pi/2$ . When  $s = 1/2$ , one has  $q = b = 0$ ,  $R = 2$  and  $\alpha = \pi/2$ ; the three circle centers coincide at  $(0, 1)$ , and the bottom segment reduces to  $(0, -1)$ . Thus

$$E_{1/2} = \{(x, y): x^2 + (y - 1)^2 \leq 4\}. \quad (2.10)$$

This is a disk, not a pinched region.

LEAN REFERENCES

```
CMVGeometry.lean volume_upper_circular_segment
TypeThreeAssembly.lean TypeThreeAssembly.core_weightedArea_formula;
                       TypeThreeAssembly.weightedArea_formula
CMVElementaryGeometry.lean CMVElementary.GeometryBridge.E_half_disk
```

For completeness, the coordinate formulas in (2.3) and (2.4) are identified exactly with the cap-and-core constructions used to prove the area and frontier formulas. Their closedness follows from the coordinate inequalities, and boundedness from their finite rectangle-and-disk decompositions. Each frontier is a finite union of circular arcs and, for  $E_s$ , the bottom segment. Exact arc-length formulas give finite frontier measure; boundedness of the density then gives both integrability assertions. Thus (2.5) consists of proved integral identities for specified sets, not assumptions supplied to the comparison.  $\square$

LEAN REFERENCES

```
CMVElementaryGeometry.lean CMVElementary.GeometryBridge.C_assembly;
                             CMVElementary.GeometryBridge.E_assembly;
                             CMVElementary.GeometryBridge.four_regularity;
                             CMVElementary.GeometryBridge.three_regularity
```

### 3. TWO COMPARISONS AT THE SAME CURVATURE

Define

$$\begin{aligned} S(z) &= B(z) - zD(z), \\ J(u) &= 2S(u) - S(2u - 1) - \pi/2, \\ j(u) &= 4(D(2u - 1) - D(u)), \quad H(u) = J(u) - uj(u). \end{aligned} \quad (3.1)$$

Here  $0 \leq u \leq 1$ . We write  $j$  for the continuous extension of  $J'$ ; it is defined by the displayed formula even at the endpoints, where no differentiation is intended.

LEAN REFERENCES

CMVElementary.lean CMVElementary.S; CMVElementary.J; CMVElementary.jprime; CMVElementary.H

**Lemma 1.** For  $\lambda > 1$  and  $0 \leq u \leq 1$ ,

$$J(u) \geq \gamma_\lambda > 0, \quad H(u) \geq \gamma_\lambda. \quad (3.2)$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.J\_lower; CMVElementary.H\_lower; CMVElementary.gamma\_pos

*Proof.* The elementary normalization

$$\frac{\sqrt{\lambda^2 - z^2}}{\lambda} = \sqrt{1 - z^2/\lambda^2} \quad (\lambda > 0)$$

identifies (2.1) with the radical normalization used in the underlying calculus. For  $-1 < z < 1$ , differentiation gives

$$\begin{aligned} D'(z) &= z \left( \frac{1}{\sqrt{1-z^2}} - \frac{1}{\lambda\sqrt{\lambda^2-z^2}} \right), \\ B'(z) + D(z) &= zD'(z), \quad S'(z) = -2D(z). \end{aligned} \quad (3.3)$$

The coefficient of  $z$  in  $D'(z)$  is positive, since  $\lambda\sqrt{\lambda^2-z^2} > \sqrt{1-z^2} > 0$ . Consequently  $D$  is even and strictly increasing on  $[0, 1]$ , with endpoint continuity supplying the closed interval.

LEAN REFERENCES

CMVElementary.lean CMVElementary.DerivativePilot.sqrt\_normalization;  
CMVElementary.DerivativePilot.hasDerivAt\_D; CMVElementary.hasDerivAt\_S;  
CMVElementary.D\_even; CMVElementary.D\_strictMono

The inverse trigonometric reflection identities yield

$$S(-z) = \lambda\pi - S(z), \quad J(0) = J(1) = S(1) - \pi/2 = \gamma_\lambda. \quad (3.4)$$

To prove positivity, put  $\theta = \arccos(1/\lambda) \in (0, \pi/2)$ . Then  $\gamma_\lambda = \lambda\theta - \sin\theta > 0$ , since  $\sin\theta \leq \theta < \lambda\theta$ . On  $(0, 1)$ , (3.3) gives  $J' = j$ . Evenness and strict increase of  $D$  show that

$$j(u) \begin{cases} > 0, & 0 < u < 1/3, \\ = 0, & u = 1/3, \\ < 0, & 1/3 < u < 1. \end{cases} \quad (3.5)$$

Indeed, compare  $|2u - 1|$  with  $u$ . Thus  $J$  increases to  $1/3$  and decreases thereafter, proving its lower bound from the equal endpoint values. On  $[1/3, 1]$ , one also has  $j \leq 0$ , so  $H \geq J$ .

LEAN REFERENCES

`CMVElementary.lean` `CMVElementary.S_reflection`; `CMVElementary.DerivativePilot.J_zero`;  
`CMVElementary.J_one`; `CMVElementary.hasDerivAt_J`; `CMVElementary.jprime_pos`;  
`CMVElementary.jprime_neg`; `CMVElementary.H_ge_J`

It remains to consider  $0 < u \leq 1/3$ . By evenness,

$$j(t) = 4(D(1 - 2t) - D(t)), \quad 0 \leq t \leq 1/3.$$

As  $t$  increases, the first argument decreases and the second increases. Hence  $j$  is nonincreasing on this closed interval. The mean value theorem, applied to  $J$  on  $[0, u]$ , supplies  $c \in (0, u)$  with

$$J(u) - J(0) = uj(c) \geq uj(u). \quad (3.6)$$

Therefore  $H(u) \geq J(0) = \gamma_\lambda$ . At  $u = 0$  the result is immediate. This is the decreasing-derivative tangent argument used for the formal lower bound: the left endpoint lies below the tangent at the right endpoint. It requires continuity of  $J$  at zero, not differentiation there.  $\square$

LEAN REFERENCES

`CMVElementary.lean` `CMVElementary.MonotoneTangent.jprime_antitone_lower_third`;  
`CMVElementary.MonotoneTangent.endpoint_tangent`;  
`CMVElementary.MonotoneTangent.h_lower_third_of_monotone_tangent`

The tangent argument is the same concavity mechanism as in [2]. The formal development also establishes, separately, all of the derivative and limit identities appearing there:

$$\begin{aligned} j'(u) &= 8D'(2u - 1) - 4D'(u) < 0, \\ H'(u) &= -uj'(u) > 0 \quad (0 < u < 1/3), \\ \lim_{u \downarrow 0} H(u) &= \gamma_\lambda. \end{aligned} \quad (3.7)$$

The signs follow because  $2u - 1 < 0 < u$ . These facts justify the alternative proof by increasing  $H$  from its limit, but that is not the lemma chain used to prove the lower-third bound above. All endpoint values of  $j$  and  $H$  are continuous extensions; no derivative of a square root at its endpoint is used.

LEAN REFERENCES

`CMVElementary.lean` `CMVElementary.auxiliary_claims`

**Proposition 2 (same-curvature comparisons).** *For  $\lambda > 1$  and  $0 < h \leq 1$ ,*

$$\begin{aligned} A_4(h) - A_3(h) &\geq \frac{\gamma_\lambda}{h^2} > 0, \\ (P_4(h) - hA_4(h)) - (P_3(h) - hA_3(h)) &\geq \frac{\gamma_\lambda}{h} > 0. \end{aligned} \quad (3.8)$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.same\_curvature\_claims

*Proof.* Substitution into (2.2), with  $q = 2h - 1$ , gives

$$\begin{aligned} h(P_4(h) - hA_4(h)) &= 2S(h), \\ h(P_3(h) - hA_3(h)) &= S(q) + \pi/2, \\ h^2(A_4(h) - A_3(h)) &= J(h) - hj(h) = H(h). \end{aligned} \tag{3.9}$$

Subtract the first two identities and apply Lemma 1; the last identity gives the area inequality. Division by the positive quantities  $h$  and  $h^2$  proves (3.8).  $\square$

LEAN REFERENCES

CMVElementary.lean CMVElementary.support\_identity; CMVElementary.area\_identity

#### 4. VARIATION AND THE GREATEST EQUAL-AREA ROOT

**Lemma 2.** *For  $\lambda > 1$ , the functions  $A_3$  and  $P_3$  are continuous on  $(0, 1]$ , differentiable on  $(0, 1)$ , and satisfy*

$$P_3'(s) = sA_3'(s) \quad (0 < s < 1), \quad \lim_{s \downarrow 0} A_3(s) = +\infty. \tag{4.1}$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.three\_arc\_calculus\_claims

*Proof.* Continuity follows from (2.1)–(2.2). On the open interval all differentiated radicals have positive arguments. With  $q = 2s - 1$ , write

$$\begin{aligned} N(s) &= B(q) + \pi/2 + (2s + 1)D(q), \\ M(s) &= 2(B(q) + \pi/2 + D(q)). \end{aligned}$$

Then  $A_3 = N/s^2$  and  $P_3 = M/s$ . By (3.3),

$$N'(s) = 8sD'(q), \quad M'(s) = 8sD'(q) - 4D(q), \quad 2N(s) - M(s) = 4sD(q).$$

The quotient rule now gives

$$s^2(P_3'(s) - sA_3'(s)) = s(M'(s) - N'(s)) + 2N(s) - M(s) = 0.$$

The reused calculus formulas are identified with (2.2) by the positive- $\lambda$  radical normalization, so they give precisely this variational identity, not a different family.

LEAN REFERENCES

CMVElementary.lean CMVElementary.scalar\_identification\_claims;  
CMVElementary.three\_arc\_variational\_hasDerivAt

Finally, as  $s \downarrow 0$ , the numerator  $N(s)$  tends to

$$B(-1) + \pi/2 + D(-1) = \lambda \arccos(-1/\lambda) + \sqrt{1 - \lambda^{-2}} > 0.$$

Division by  $s^2 \downarrow 0$  proves the divergence.  $\square$

LEAN REFERENCES

CMVElementary.lean CMVElementary.A3\_tendsto\_atRight\_zero

**Lemma 3 (greatest equal-area root).** *Let  $\lambda > 1$  and  $0 < h \leq 1$ , and put  $V = A_4(h)$ . There exist  $0 < \varepsilon < r < h$  such that the set*

$$Z = \{s \in [\varepsilon, h] : A_3(s) = V\} \quad (4.2)$$

*is nonempty and compact,  $r = \max Z$ , and*

$$A_3(r) = V, \quad A_3(s) < V \quad (r < s \leq h). \quad (4.3)$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.Z; CMVElementary.greatest\_root\_claims

*Proof.* Proposition 2 gives  $A_3(h) < V$ . By Lemma 2, choose  $0 < \varepsilon < h$  with  $A_3(\varepsilon) > V$ . The intermediate value theorem gives a point of  $Z$ . Continuity makes  $Z$  a closed subset of the compact interval  $[\varepsilon, h]$ , so it has a greatest element  $r$ . The strict endpoint inequalities give  $\varepsilon < r < h$ . If  $r < s \leq h$  and  $A_3(s) = V$ , maximality fails. If  $A_3(s) > V$ , continuity between  $s$  and  $h$  gives a later equal-area root, again a contradiction. This proves (4.3). No root uniqueness or global monotonicity of  $A_3$  has been assumed.  $\square$

LEAN REFERENCES

GreatestLevelRoot.lean CMVElementary.exists\_greatest\_level\_root

## 5. THE EQUAL-AREA PERIMETER COMPARISON

**Theorem 2 (scalar comparison).** *For every  $\lambda > 1$ , one has  $\gamma_\lambda > 0$ ; for every  $0 < h \leq 1$ , there exists  $0 < r < h$  such that*

$$A_3(r) = A_4(h), \quad P_4(h) - P_3(r) > \frac{\gamma_\lambda}{h} > 0. \quad (5.1)$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.scalar\_comparison

*Proof.* Choose the greatest root from Lemma 3 and define

$$Q(s) = P_3(s) + s(V - A_3(s)), \quad r \leq s \leq h. \quad (5.2)$$

This function is continuous on  $[r, h]$ . On its interior, Lemma 2 and the strict area barrier give

$$Q'(s) = P_3'(s) + V - A_3(s) - sA_3'(s) = V - A_3(s) > 0. \quad (5.3)$$

The mean value theorem therefore gives the strict comparison

$$P_3(r) = Q(r) < Q(h) = P_3(h) + h(A_4(h) - A_3(h)). \quad (5.4)$$

LEAN REFERENCES

CMVElementary.lean CMVElementary.Q; CMVElementary.greatest\_root\_claims  
GreatestLevelRoot.lean CMVElementary.strict\_support\_comparison

Subtracting (5.4) from  $P_4(h)$  and using Proposition 2 yields

$$P_4(h) - P_3(r) > \frac{\gamma_\lambda}{h} > 0. \quad (5.5)$$

This proves the scalar theorem. The complete elementary route consists of the auxiliary bounds, the two same-curvature identities, the variational calculus and its formula identifications, and this greatest-root construction with the strict  $Q$  comparison; all follow from  $\lambda > 1$ .  $\square$

LEAN REFERENCES

CMVElementary.lean CMVElementary.elementary\_route

*Proof of Theorem 1.* The root in Theorem 2 satisfies  $0 < r < h \leq 1$ , hence  $r < 1$ . Proposition 1 applies to  $C_h$  and  $E_r$ . Substituting their actual area and perimeter identities into (5.1) gives (1.3), with exact equality of weighted areas and a strictly positive perimeter improvement.  $\square$

LEAN REFERENCES

CMVElementaryGeometry.lean CMVElementary.geometric\_comparison

## 6. ENDPOINTS AND SCOPE

The endpoint  $h = 1$  causes no loss: the four-arc coordinate construction remains valid, the scalar formulas are continuous at one, and differentiation in the final comparison takes place only on  $(r, h)$ . The competitor remains strictly inside the three-arc parameter interval. Major caps and the radius-two disk are already included in Proposition 1. The choice of a greatest root, rather than a presumed unique root, is essential to the stated argument.

The improvement is positive for every fixed  $\lambda > 1$ , but no positive margin uniform as  $\lambda \downarrow 1$  is claimed. Formula (1.2) gives  $\gamma_\lambda \rightarrow 0$ . Thus the phrase “for every density contrast” does not mean a density-independent lower bound.

**Corollary (informal; using the external classification).** *For each  $\lambda > 1$ , there is a transition area  $v_0 > \pi$  with the ball, stadium and three-arc alternatives of [1, Conjecture 3.12]. For  $0 < v < \pi$ , minimizers are disks contained in the strip; for  $\pi < v < v_0$ , they are stadiums; for  $v > v_0$ , they are three-arc regions. At  $v = \pi$  the stadium degenerates to a disk. At  $v = v_0$ , stadium and three-arc minimizers coexist. The classification is up to density-preserving isometries and the relevant measure equivalence; no uniqueness is asserted.*

*Proof using [1].* Proposition 3.9 supplies vertical reflection symmetry for arbitrary minimizers. Lemma 3.8 then supplies its four candidate families; Step 2 of its proof gives horizontal symmetry of the four-arc family, placing that family, after a density-preserving isometry, in the form  $C_h$ . Theorem 1 excludes it. Theorem 3.16 of [1] supplies the remaining area ranges and transition coexistence, so its residual four-arc alternative disappears.  $\square$

This corollary uses a further measure-theoretic transfer. For the explicit finite piecewise-circular Jordan regions here, the complete and reduced boundaries agree up to an  $\mathcal{H}^1$ -null set, so their weighted boundary integrals agree, including the density-one interfaces. This agreement was already stated in [2, §3]; it was not omitted from that draft. For an arbitrary minimizer, one must first invoke the external regular-representative and classification results. Reduced-boundary perimeter is insensitive to changes on sets of planar measure zero; complete-frontier perimeter need not be. Accordingly, [2]’s broad conjecture claim is justified as the cited corollary, while its final explicit-family scope remark describes the self-contained theorem.

The Lean-verified results are the explicit geometric comparison, the separately named scalar comparison, the elementary route, and the unconditional geometric realization. The six frozen lower-level modules supply calculus, circular geometry, and integral/frontier infrastructure, not an earlier final exclusion theorem. Neither the external classification nor the bounded-variation and reduced-boundary transfer is formalized or introduced as an extra axiom. Exhaustive declaration mappings, source code, and retained verification evidence belong to the supplement rather than the mathematical argument.

#### REFERENCES

- [1] Antonio Cañete, Michele Miranda Jr. and Davide Vittone, *Some isoperimetric problems in planes with density*, Journal of Geometric Analysis **20** (2010), 243–290. Preprint arXiv:0906.1256. In particular, §2, Lemma 3.8, Proposition 3.9, Conjecture 3.12 and Theorem 3.16.
- [2] Skilyn Leon, Rosa Pavlak, Evelyn Pulla and Xi Sisi Shen, *An elementary comparison for four-arc regions in a planar strip density*, supplied undated note, `CMV_Strip_Density_Problem.pdf`, §§3–6 and Remarks 1–3; the two printed Theorem 1 inequalities are corrected in (1.3).